

The Cost of Not Being Seen:

Automated Licence-Plate Reader Exposure on American Commutes and the Price of Avoiding It

TRAMES Project

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Abstract

Automated licence-plate readers (ALPR) have moved from tollbooths to residential intersections, and a single vendor—Flock Safety—now accounts for 80.0 % of the 142,991 installations mapped in North America. Public argument about this technology is conducted almost entirely in the abstract. We supply a measurement instead: how much ALPR surveillance a typical American commuter actually drives through, what it would cost them to refuse it, and how that answer has changed as the map of cameras filled in.

We build a continental routing engine that models ALPR fields of view as directional geometry—98.0 % of mapped cameras carry a bearing—and can be asked for the fastest route that crosses none of them. We route 56,131 real home–work commutes, drawn from Census LEHD LODES data in all fifty states and the District of Columbia and weighted to each state’s commuters, twice each on an identical road network.

Three findings. First, 75.3 % of commutes pass at least one ALPR, a median of three and a 99th percentile of thirty-four. Second, refusal is far cheaper than expected: 84.2 % of commutes can be routed to *zero* ALPR exposure for a median of 1.62 min, an overhead of 6.71 %. Third, and most consequentially for policy, demographic gradients that appear robust nationally—origin tracts in the highest quartile of Hispanic population pass $1.46\times$ as many cameras as those in the lowest, and of Black population $1.39\times$, with intervals well clear of parity—vanish when tracts are ranked within their own county ($1.01\times$ and $1.06\times$, indistinguishable). Mapped camera density varies 7.1-fold per capita between the 10th and 90th percentile states. We therefore report the equity result as a between-county phenomenon that this design cannot attribute to siting, mapping effort, or urban form, and we argue that no study built on crowdsourced surveillance data should report such a gradient without this check.

Resolving exposure by vendor sharpens the picture: tolling and transport-agency cameras show a national gradient of $1.57\times$ against $1.34\times$ for Flock, and per unit deployed reach $2.3\times$ as many commuters. Flock’s gradient dissolves under within-county ranking; the transport-agency gradient narrows to $1.16\times$ and sits exactly on the line.

Finally, we reconstruct the camera map as it stood on five dates since January 2024 from OpenStreetMap’s edit history—exactly, node for node—and route the same commutes against each. The map grew 128-fold, and measured exposure almost in proportion, from 2.2 % of commuters passing a mapped reader to 75.3 %. Refusing them grew dearer as it did, per camera as well as in total—for commuters with a camera to avoid, from a median of 0.8 min to 3.1 min—partly because the detours were being mapped too. The national equity gradients moved with the map—the income contrast was $1.58\times$ in January 2025 and is $1.05\times$ today—while the within-county ones held at parity or converged toward it.

All code, data acquisition, and analysis are reproducible from public sources.

1 Introduction

Automated licence-plate readers photograph every passing vehicle, resolve the plate, and retain the read. Unlike a traffic camera they are not triggered by an offence, and unlike a patrol officer they do not tire. Fixed, pole-mounted units have spread from limited-access highways into residential neighbourhoods, arterial roads, and the entrances of private subdivisions.

The public debate about this expansion is conducted in the vocabulary of policy: retention windows, warrant requirements, audit logs, data-sharing agreements. What that vocabulary lacks is a measurement of the thing being regulated. How much of this does a person actually drive through on an ordinary Tuesday? And if they would rather not, is declining a practical option or a theoretical one?

This paper answers both, and does so by building the instrument the question implies. We construct a routing engine that treats ALPR fields of view as first-class geometry and can be asked for the fastest route that avoids them. Running each sampled commute through it twice—once normally, once avoiding—yields the exposure and the price of refusing it, for the same trip, on the same network, under the same traffic model.

The second quantity is where the paper earns its keep. A privacy protection that costs four minutes is a feature. One that costs forty is available only to people with forty minutes to spare, and a right that is priced is not evenly held. We therefore report the cost of refusal across origin-tract income and racial composition—subject to the substantial caveat developed in Section 7.2 and quantified in Section 5.5, which turns out to govern how the demographic results may be read.

A third question follows from the data source. The cameras are mapped by volunteers, and the map has grown 128-fold since January 2024. OpenStreetMap keeps every edit, so the measurement can be repeated on the map as it stood on any past date; we repeat it on five. How much surveillance a commuter drives through turns out to be, in large part, a question of how much of it had been mapped—and that has consequences for every equity claim made from such data.

1.1 Contributions

- **A directional exposure model at continental scale.** Cameras are wedges, not circles. 98.0% of mapped North American ALPR installations carry a bearing tag, making this feasible nationally. Proximity-based counting inflates exposure by charging drivers for cameras aimed at the opposite carriageway; we do not do it.
- **Avoidance routing with no forked router.** 135,210 camera cones are compiled into a stock GraphHopper spatial index at import time, so an ordinary per-request custom model can reference them by name. No engine patches, no geometry in the request.
- **A commute-weighted rather than area-weighted sample.** 56,131 real home–work pairs in all fifty states and the District of Columbia, drawn from LEHD LODES with probability proportional to worker count and weighted to each state’s commuters.
- **A vendor-resolved account of exposure.** Flock and the DOT/tolling vendors follow different deployment logics—who *buys* surveillance versus where road infrastructure sits. Separating them shows the national racial gradient is substantially stronger in transport infrastructure than in the vendor that dominates both the network and the public argument about it.
- **A coverage-bias control that changes the conclusion.** We show that the national demographic gradients do not survive within-county ranking, quantify the 7.1-fold per-capita spread in mapping intensity that plausibly drives them, and argue this check should be mandatory for work of this kind.

- **A thirty-three-month trend on a fixed sample.** The camera map, reconstructed exactly from OpenStreetMap’s edit history on five dates since January 2024, with the same commutes routed against each: exposure rose almost in proportion to the map, refusal grew dearer per camera as well as in total, and the national equity gradients moved with the map while the within-county ones did not.

2 System

2.1 Why a fork, and of what

The measurement needs a router that treats “avoid these 135,210 polygons” as a cost rather than a post-hoc filter, and the eventual application needs to be a real navigation client rather than a research script.

OsmAnd (GPLv3) is the base: the only mature Android navigation client with first-class OpenStreetMap support, offline maps, and a documented online-routing extension point. Its *code* is GPLv3; its *artwork* is CC-BY-NC-ND 4.0 and must therefore be replaced rather than modified in a derivative.

GraphHopper 11.0 is the routing engine. Valhalla was evaluated and rejected: its `exclude_polygons` support carries open correctness defects (#3266, #4659, #5069) whose failure mode is silently routing *through* the excluded region—which is precisely the measurement everything here depends on.

One architectural property made the rest tractable. OsmAnd’s online routing engines return *geometry only*; turn instructions are re-derived client-side from offline map data. A custom online router therefore never has to synthesise navigation instructions.

2.2 The keystone: import-time custom areas

GraphHopper resolves `custom_areas.directory` into a spatial index *at graph import time*. We verified empirically—the documentation is ambiguous—that an area registered this way can be referenced by name from a *per-request* custom model:

```
{ "priority": [ { "if": "in_alpr", "multiply_by": 0.01 } ] }
```

This is what makes the design possible without forking the engine; shipping 135,210 polygons in every routing request is not viable. The cost is that regenerating cones requires a full re-import (82 min to 113 min for North America, by load).

Two further configuration decisions are load-bearing:

- **Contraction Hierarchies are disabled.** CH bakes edge weights into the prepared graph, which is incompatible with a per-request avoidance strength. Landmarks (LM) are used instead.
- **The `car_alpr` profile shares car’s LM preparation.** This halves preparation time and is valid *only* because avoidance multiplies priority strictly downward, so the borrowing profile’s weights never exceed the preparation profile’s on any edge—the condition LM correctness requires.

2.3 Deployment posture

The routing server binds to localhost. It has no authentication and no rate limiting and is not internet-facing. Any public deployment must sit behind a reverse proxy supplying both.

Table 1: Directional and vendor tagging across the two 2026 snapshots. The direction-key rows overlap (392 and 415 nodes carry both keys). Vendors are matched as case-insensitive substrings under any of the four vendor keys, in both columns.

Property	2026-07-24		2026-09-22	
	Count	Share	Count	Share
Camera nodes	120,838		142,991	
Any direction tag	119,249	98.7 %	140,112	98.0 %
via <code>direction</code>	117,356	97.1 %	138,216	96.7 %
via <code>camera:direction</code>	2,285	1.9 %	2,311	1.6 %
Multi-head (;-separated)	7,014	5.8 %	9,150	6.4 %
Arc-range form (199-269)	4,806 tokens	—	4,632 tokens	—
<code>operator</code> tagged	22,354	18.5 %	23,872	16.7 %
Flock Safety	99,650	82.5 %	114,360	80.0 %
Motorola Solutions	5,993	5.0 %	7,411	5.2 %
Genetec	2,876	2.4 %	3,721	2.6 %
Leonardo	1,020	0.8 %	1,215	0.8 %
Axis Communications	1,052	0.9 %	3,327	2.3 %
No vendor tag	5,811	4.8 %	6,850	4.8 %

3 Data

All sources are public and key-free.

3.1 ALPR installations

Source: OpenStreetMap, queried through the Overpass API for nodes tagged `man_made=surveillance` with `surveillance:type=ALPR`, matched case-insensitively (19 readers are tagged `alpr`)—the corpus surfaced by the DeFlock project, taken from the upstream rather than a mirror. Retrieved in five regional queries (continental US, Alaska, Hawaii, Canada, Mexico) and deduplicated by OSM node id. The results in this paper rest on the snapshot of **2026-09-22: 142,991 unique camera nodes**, 142,257 of them in the 50 states and the District of Columbia (Figure 1); the rest fall in the parts of Canada and Mexico the regional queries cover. A first snapshot, taken 2026-07-24, held 120,838 (120,272 in the states), and Section 3.3 reconstructs five earlier dates.

A fetch that fails over between Overpass mirrors has to check the mirror’s database date. On the first attempt at the September snapshot one regional query fell through to a mirror whose database was dated 2026-05-06; that box overlaps the southern United States, regions were merged last-wins, and roughly 13,600 cameras across Texas, Arizona and Florida silently reverted to four-month-old tags while some 400 nodes deleted since May returned as ghosts. Nothing failed. The fetch now rejects any answer older than 48 hours and merges overlapping regions newest-database-first; every tile of the snapshot used here falls within a 13-minute window.

Three tagging realities shaped the parser, each contradicting a plausible assumption:

1. **direction, not camera:direction.** The OSM wiki documents `camera:direction` for surveillance nodes. Mappers overwhelmingly use plain `direction`, by a factor of 60. Reading only the documented key discards 97 % of the bearings.
2. **Arc ranges are real, and mostly 45°.** 4,632 values give an explicit range; 3,806 of them

(a) 142,257 mapped ALPR installations, by vendor

(b) Installation density (log scale)

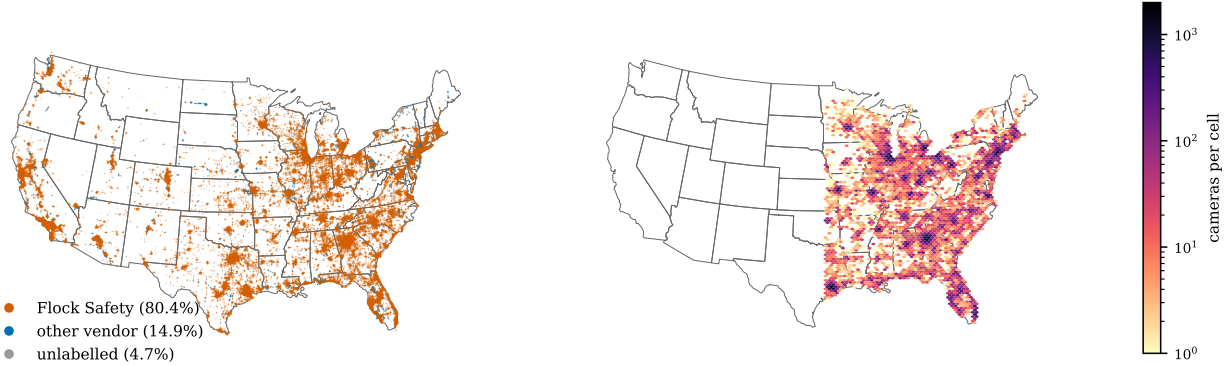


Figure 1: Mapped ALPR installations in the contiguous United States. (a) By vendor: Flock Safety dominates by count almost everywhere, with other vendors concentrated on limited-access corridors. (b) Density on a log scale. The visible structure is partly deployment and partly *mapping*: see Section 5.5.

Table 2: Vendor-split cone sets, 2026-09-22 snapshot. The cones-per-camera column is itself a result: non-Flock installations are multi-head more than six times as often (22.8% against 3.6%), consistent with tolling and DOT gantries covering every lane while a Flock unit watches one approach.

Set	Cameras	Cones	Cones/camera	Union parts
Flock Safety	113,611	120,366	1.06	109,427
Other named vendor	20,435	30,128	1.47	21,587
No vendor tag	5,997	6,590	1.10	5,211

(82%) span exactly 45° . That is what justifies 45° as the default span for the $\approx 97\%$ of cameras giving a bare bearing—the default is inherited from the data, not assumed.

3. **Multi-head units exist** (320;190, 0;72;144;216;288, 199-269;325-35); 9,150 nodes. These typically watch *both* carriageways, so silently taking the first bearing is the worst available error: it makes a camera that sees everyone look like one that sees half. The parser emits one wedge per head.

3.2 Cone construction

Each camera head becomes a circular sector: apex at the node, bisector along the tagged bearing, span 45° (or the tagged arc), radius 60 m. The radius must reach across the carriageway from a set-back pole; it was chosen on that reasoning and then swept over $\{30, 45, 60, 90\}$ m (Section 7.3). Measured exposure is only mildly sensitive to it, but the avoidance result is not robust above the assumed value.

All wedges are unioned into a single MultiPolygon of **135,210 parts**, published as one feature with id `alpr`. Overlapping wedges merging is correct and intended: the question is *is this road segment observed*, not *by how many devices*.

Assigning vendor requires care. Vendor appears under four keys (`manufacturer`, `surveillance:manufacturer`, `brand`, `surveillance:brand`) and is spelled inconsistently. An exact match on `manufacturer = "Flock Safety"` finds 109,892 cameras; a case-insensitive substring search across

all four finds 114,360. The 4,468 difference is mostly 4,711 nodes carrying the vendor under **brand**, plus variants (“Flock Group Inc.”, “FlockSafety”). Misfiling 3.9 % of Flock cameras as non-Flock is the one error a vendor comparison cannot absorb.

Untagged cameras are held out of *both* vendor sets rather than assigned to the complement of Flock, since an untagged camera is likelier to be an unlabelled Flock unit than a non-Flock one. They are reported as their own set so nothing is silently dropped, with the caveat that their direction coverage is lower (88.0 % against 99.4 % for Flock and 93.8 % for other named vendors—the last down from 98.5 % in July, as newly mapped Axis units often arrive without a bearing).

3.3 Camera maps of earlier dates

OpenStreetMap keeps every edit, so the camera map can be reconstructed as it stood on any past date. The public Overpass server answers a query carrying a `[date:...]` qualifier from its history: nodes deleted since are restored, nodes added since are absent, and every tag has its value from that day. Asked for Georgia at the July snapshot’s timestamp it returned 9,742 nodes, the snapshot’s own count to the node. History queries for the whole continent exceed the server’s fifteen-minute limit, so a date is fetched state by state and merged by node identifier; DeFlock’s own Overpass mirror keeps no history and cannot serve them. They are heavy, and after one complete date the public server began refusing our connections.

The remaining dates were therefore reconstructed locally, from OpenStreetMap’s full-history dump (163 GB, 14 September 2026, checked against its published MD5), with Overpass’s semantics: every version of every node that was ever tagged as an ALPR, including the later versions that deleted or retagged it, and for each date the latest version at or before midnight UTC, kept if it was visible and still tagged. On the one date both sources cover, 1 July 2025, they agree exactly—22,690 nodes, the same identifiers, the same tags, the same coordinates.

We reconstruct five dates—1 January and 1 July of 2024 and of 2025, and 1 January 2026—and set them beside the study’s July and September 2026 snapshots, seven maps in all. Cones are built from each exactly as from the current map (Section 3.2). The share of mapped cameras carrying a bearing, and so receiving a cone, was about 90 % in 2024 and 98 % in 2026: the earliest maps are slightly less complete even on their own terms. Monthly counts between the dates come from the ohsome history API, which aggregates but will not release individual records; it agrees with our July snapshot to 21 nodes, exactly the readers tagged lowercase `alpr` that the July fetch’s case-sensitive match omitted.

One caveat governs everything built on these maps. OpenStreetMap records when a camera was *mapped*, not when it was installed: an installation date is present on 47 of 142,991 nodes. A camera first mapped in 2026 may have been on its pole since 2021. A past map is therefore what a study run on that day would have seen, and the trend of Section 5.7 describes how the measured picture changed as volunteers filled in the map—most of it after DeFlock was founded in October 2024¹—not how fast cameras were deployed.

3.4 Road network

Geofabrik North America extract, 18 GB, retrieved 2026-07-24 and held fixed for every camera map, so that the trend of Section 5.7 isolates the camera data: every unavaoided route is the same route on every date. Imported with `ignored_highways = footway,construction,cycleway,path,steps`. `track` is deliberately retained: unpaved tracks are plausible avoidance routes, and excluding them would bias the cost estimate downward by removing options a determined driver has.

¹<https://deflock.me>; founding date per <https://deflockilm.org/what-is-deflock/>.

3.5 Commute sample

Source: LEHD **LODES8**, `od_main_JT00`—all jobs, both ends in-state, built from unemployment-insurance wage records linked to employer establishments. These are administrative counts of real jobs, not survey estimates. The year is 2022 for 49 of the 51 jurisdictions; LODES has not published 2022 for Michigan, which uses 2021, or for Alaska, whose latest year is 2016. LODES8 codes every year against 2020 census geography, so an older year still names current tracts.

Sampling design carries more weight here than sample size. Uniform random points would over-represent empty rural space where nobody commutes and no cameras exist, understating exposure badly. Instead each origin–destination pair is drawn **with probability proportional to the number of workers actually making that trip**, so the distribution approximates what an average *commuter* experiences rather than what an average *square kilometre* contains.

Block-level LODES geocodes are aggregated to **tract** (first 11 digits)—the finest resolution at which ACS demographics are reliable, and finer than the routing’s effective precision. Same-tract commutes are dropped: origin and destination centroids coincide, so there is no route to measure. Georgia alone yields 1,199,250 distinct tract pairs covering 4,190,313 workers; across the 50 states and the District of Columbia the frame holds 132,822,679 commuters over 34,420,305 tract pairs.

Connecticut needed one repair. In 2022 it replaced its eight counties with nine planning regions, and every tract identifier changed with them. LODES still names tracts by the old counties while the 2024 Gazetteer and ACS 2022 use the new ones, so without a crosswalk no Connecticut tract matches and the state silently drops out of the sample. The tracts were reassigned, not redrawn: each six-digit tract code maps to exactly one new identifier, and all 879 in the flows do.

1,200 pairs are drawn per jurisdiction, 61,200 in all, with replacement from a single seeded stream (seed 20260725), giving **58,421 distinct commutes**. The fifteen states of the study’s earlier rounds are drawn first, so their 17,904 commutes are reproduced exactly and their routes carry over. The sampled pairs represent 2,842,747 workers; the median pair carries 16 and the largest 2,832.

A pair drawn more than once is routed once, since its route is the same, but it stands for every draw: **each commute carries its draw count as a weight**. In the fifteen large states this is immaterial—0.53% of draws are repeats—but in small ones it is not. Wyoming’s 1,200 draws hold 832 distinct pairs (30.7% repeats), Alaska’s and the Dakotas’ repeat a fifth of the time or more, and one Vermont pair was drawn nine times. Collapsing repeats at equal weight, as this study’s earlier rounds did, under-represents precisely the heaviest flows; keeping the counts restores probability-proportional-to-workers sampling inside every state.

3.6 Demographics and geography

Tract centroids are 2024 Census Gazetteer internal points (`INTPTLAT/INTPTLONG`), which are guaranteed to fall inside the tract polygon. Income is ACS 5-year 2022 table **B19013**; race and ethnicity are table **B03002**, chosen over B02001 because it is Hispanic-aware, so “non-Hispanic Black” is directly available rather than reconstructed. Demographics attach to the *origin* (home) tract: the claim under test concerns who lives where surveillance is dense.

4 Method

4.1 Two routes per commute

Each sampled pair is routed twice against the same graph: a **baseline** with no custom model, and an **avoided** route applying priority: `[{if: "in_alpr", multiply_by: 0.01}]`. Both

disable CH, which per-request custom models require.

Both use GraphHopper’s stock car model, which weighs distance as well as time (`distance_influence` 90: a route 1km shorter is preferred if it costs under 90s more). The baseline is therefore the router’s preferred route rather than strictly the fastest, and for 4.5 % of commuters the avoiding route—in every such case the longer one—is the quicker, by a median of 0.9 min. Extra time is reported as measured, negative values included; clipping them at zero leaves every median unchanged and raises the mean extra time from 5.62 min to 5.70 min.

0.01 is the *maximum* avoidance strength, not a typical setting. Results therefore describe an **upper bound on both the benefit and the cost** of avoidance—the strongest available preference short of prohibition. This is deliberate: it establishes whether meaningful avoidance is possible at all, and what it costs at the limit.

4.2 Exposure is directional, not proximate

A camera counts as having seen the driver *only if the route line intersects its cone*. Passing 20 m from a camera aimed the other way is not exposure. Implementation is an STRtree over the 135,210 cone parts for candidate lookup, then an exact intersection test.

Two conservatisms remain, both understating exposure: route geometry is a polyline while a vehicle has width, and a cone occluded by intervening buildings is still counted as full sight-line.

4.3 Vendor attribution costs no additional routing

Routing does not change between vendor questions—the avoidance model targets the all-vendor union either way—so Section 5.6 is entirely a matter of re-scoring existing routes against different cone sets. This is possible only because the experiment persists route polylines to a sidecar; keeping only counts would have made both the vendor split and the radius sweep of Section 7.3 fresh multi-hour re-runs.

As a consistency check, the three vendor sets should sum to slightly *more* than the all-vendor union, because cones from different vendors that overlap merge into one part in the union and are counted once while the split sets count them separately. Measured excess is 1.0 %: right sign, plausible magnitude.

4.4 Earlier camera maps need no new baselines

The unavoided route does not depend on the cameras. Routed two months apart against graphs carrying different camera maps, all 17,580 commutes of the study’s earlier rounds took identical baseline routes, to the metre and the second. Exposure on an earlier date is therefore a re-scoring of each saved route against that date’s cones, with no routing.

Avoidance does need routing against that date’s cameras, and two properties keep it tractable. First, one graph serves every date: each date’s cone set is imported as its own custom area (`alpr_2025_07_01, ...`), and each request names the one it avoids—a single import of about two hours rather than one per date, checked on a test graph to route each area independently. Second, only commutes that pass a camera on that date are routed. A route that crosses no cone is already the avoiding route, since the custom model only makes cone edges dearer and nothing can undercut it; this was checked rather than assumed, and all 3,741 such commutes of the fifteen-state run received an identical avoiding route. The same shortcut serves the main run.

Re-scoring uses route geometry stored at five decimal places, about a metre. Against live scoring on the current map it moves 1.7 % of commutes by one camera, in the two directions about equally, changes whether 5 of 17,580 (0.03 %) pass any camera at all, and leaves national totals

within 0.1 %. On the full sample the re-scored national figures match the live ones to the precision reported (75.3 % passing a camera, a mean of 5.14). Within the trend every date, the current one included, is scored the same way. The July 2026 snapshot enters the trend re-scored only: routing its avoidance in all fifty-one states would have added most of a day of computation to a step that the fifteen-state paired comparison already covers (Section 5.7).

4.5 Statistical reporting

- **Medians lead, means accompany.** Both distributions are heavily right-skewed; a minority of long commutes through dense corridors pull the mean well above the typical experience.
- **National figures are weighted by commuters.** Every jurisdiction was sampled at the same size, so an unweighted pool would count Wyoming’s 198,447 in-state commuters as heavily as California’s 16,821,903. Each routed commute carries its state’s commuter total from the LODES frame, shared among that state’s routed commutes in proportion to draw count, and quartile cut-points and medians are weighted quantiles. State figures are within-state estimates and need only the draw counts. The weights matter even on the fifteen states of the earlier rounds: weighting alone moves the share of commuters passing a reader there from 78.7 % to 81.1 %, because the populous states are the more surveilled ones.
- **Confidence intervals from a bootstrap stratified by state** (percentile, 2,000 replicates, seed 20260725). Each replicate resamples every state at its own size and re-derives its weights, as the sample was drawn; a plain bootstrap over pooled rows would let one replicate hold 1,400 Texans and 900 Californians, variation the design does not have. Camera counts are discrete and zero-inflated, nowhere near normal, so no interval is assumed symmetric.
- **Quartile contrasts, not regressions.** With tract-level aggregates a regression coefficient invites a causal reading the design cannot support.
- **Total exposure and per-kilometre density are reported as peers.** A raw count gap that vanishes once normalised per km is about distance, not siting. But the converse is a finding, not a nuisance: a group whose commutes are *shorter but more densely watched* shows a rate gap and no count gap, and treating the rate purely as a control would discard it. Cameras per km is a ratio of sums bootstrapped over commutes; a mean of per-commute rates would let very short commutes dominate.
- **Ecological inference is not attempted.** Tract composition is a property of tracts, not of the individuals routed.

4.6 Execution

Twelve concurrent workers against the local routing server. Worker count was measured, not assumed: twelve completed 300 commutes in 324 s, while thirty-two took longer than 576 s on the identical set—past about twelve, contention outweighs added concurrency. Routing dominates cost overwhelmingly; the geometric exposure test is 0.2 % of per-commute work.

Failed routes are dropped rather than retried, and **56,131 of 58,421 commutes routed (3.9 % dropped)**. The failures are not spread evenly, and not mostly what they seem. Nine in ten are points the router cannot snap to any road: a tract’s internal point can lie deep inside a national park, a reservation or a wilderness tract, beyond the router’s road search. The rest are genuinely unconnected—commutes between Hawaiian islands, or to Alaskan towns off the road network. Five states lose more than a tenth of their sample (Alaska 33 %, Hawaii 24 %, Montana and Wyoming 21 %, New Mexico 14 %); the median state loses 2.0 %. Those five hold 1.6 % of the nation’s commuters, so the national figures barely register the loss, but their own estimates

Table 3: Exposure and the cost of avoidance for American commuters: 56,131 routed commutes in the 50 states and the District of Columbia, weighted to the 132.8 million in-state commuters they represent. Medians are weighted; intervals come from a bootstrap stratified by state. Median commute 23.6 km / 23.4 min.

	Median [95 % CI]	Mean
<i>Unavoided route</i>		
Cameras passed	3 [3–3]	5.14
Commuters passing ≥ 1	75.3 % [74.8–75.8]	
Commuters passing ≥ 5		36.5 %
Commuters passing ≥ 10		16.3 %
90 th / 99 th percentile	13 / 34 cameras	
Exposure rate	0.096 cameras/km	
<i>ALPR-avoiding route</i>		
Cameras passed	0 [0–0]	0.26
Reduced to zero	84.2 % [83.7–84.7]	
Extra time (min)	1.62 [1.57–1.68]	5.62
Extra distance (km)	0.57 [0.55–0.60]	2.53
Avoidance overhead	6.71 % [6.50–6.93]	13.27 %
Minutes per camera evaded	0.76 [0.75–0.78]	1.30

describe the routable part of each state—probably its less remote part, and so probably overstate its exposure. A wider road search would recover most of them, but it also moved 5 of 100 already-routed commutes to a different road (the default search sometimes stops at a road that is not the nearest), and the trend of Section 5.7 pairs every saved route with new ones. Consistency was kept over coverage.

Commutes are processed in seeded random order. The sample is grouped by state and the executor preserves input order, so in file order results arrive one state at a time: an interruption would leave some states complete and others absent, and no partial output could support a national estimate. Shuffling makes any prefix a near-uniform sample of the whole. This proved material rather than hypothetical—in the study’s first round an interim read of a single state gave a median avoidance overhead of 22.7 % against that round’s national 5.96 %.

5 Results

5.1 Exposure on the unavoided commute

75.3 % of American commuters pass at least one ALPR on the way to work. The median commuter is read three times each way; the 90th percentile thirteen times and the 99th thirty-four. Aggregate exposure is 0.096 cameras per kilometre driven. Across the fifteen states of the study’s earlier rounds the same weighted share is 81.1 %: the thirty-six states and the District added here are, on the map, far less surveilled—a difference Sections 5.3 and 5.5 will not let us attribute to the roads rather than to the map.

5.2 The cost of refusal

84.2 % of American commuters could be routed to zero ALPR exposure, at a median cost of 1.62 min and 0.57 km—an overhead of 6.71 %. The median marginal price is 0.76 min per

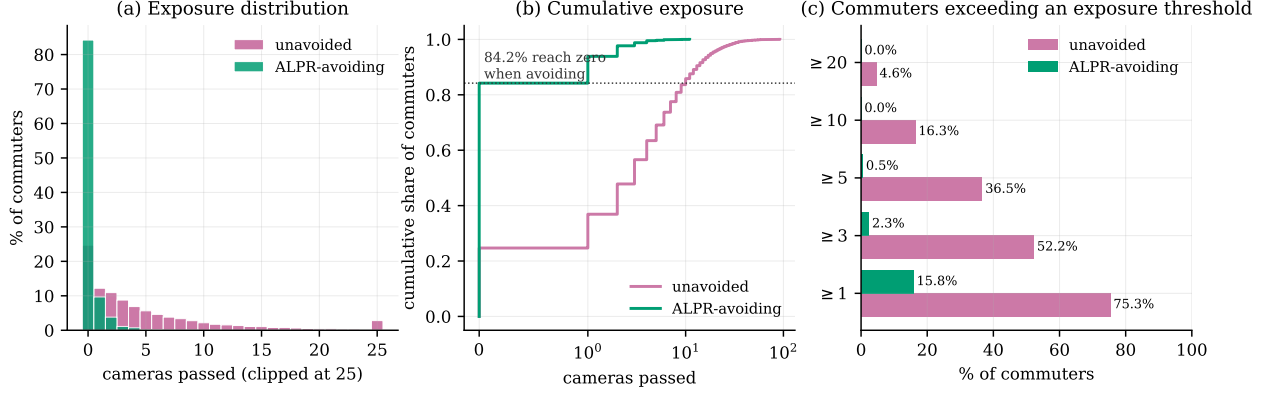


Figure 2: Exposure before and after ALPR-avoiding routing, as shares of commuters. (a) Distributions; (b) cumulative, log-scaled abscissa; (c) share of commuters exceeding a threshold. The zero-exposure mass moves from 24.7% to 84.2%.

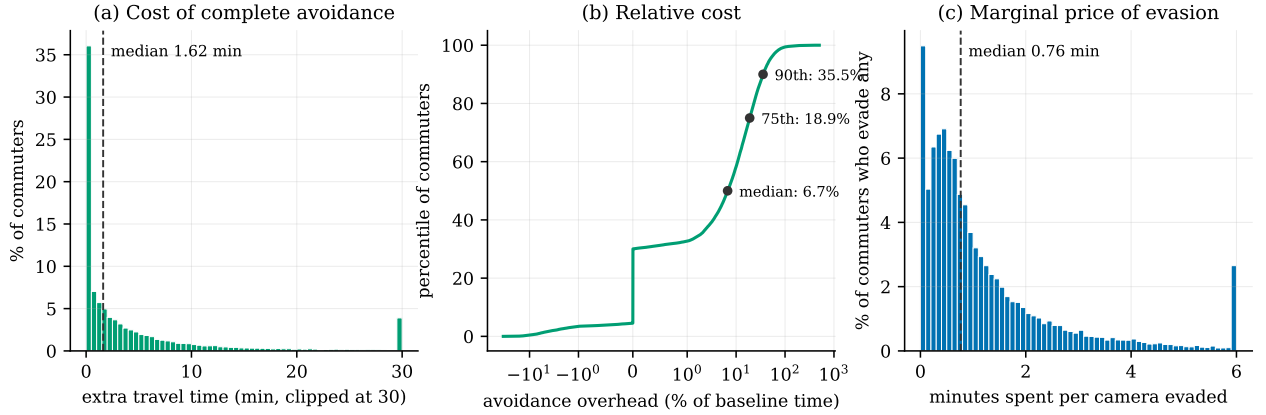


Figure 3: The price of complete avoidance, as shares of commuters. (a) Absolute extra time; (b) overhead as a percentage of baseline, by percentile; (c) marginal minutes per camera evaded. All three are heavily right-skewed—a small tail of commutes pays a great deal, while the typical commute pays almost nothing.

camera evaded.

For five commuters in six, then, complete avoidance of licence-plate surveillance costs about a minute and a half. This figure is conditional on the assumed camera field of view: it is stable if the true radius is smaller than the 60m assumed here and degrades if it is substantially larger, which Section 7.3 quantifies. It is the paper’s central quantitative claim, and it is considerably more favourable than the metropolitan spot-checks that motivated the study suggested.

The distribution matters as much as the median. The mean overhead is 13.27% against a median of 6.71%, and the interquartile range runs from 0% to 18.9%: a quarter of commuters need no detour whatsoever, while a tail pays substantially. Figure 4 shows one such case.

5.3 Fifty states and the District of Columbia

State-level variation is extreme (Figure 5; every state is tabulated in Table 9). In Georgia 91.9% of commuters pass a mapped reader, a mean of 11.8 each way, and avoiding them costs a median

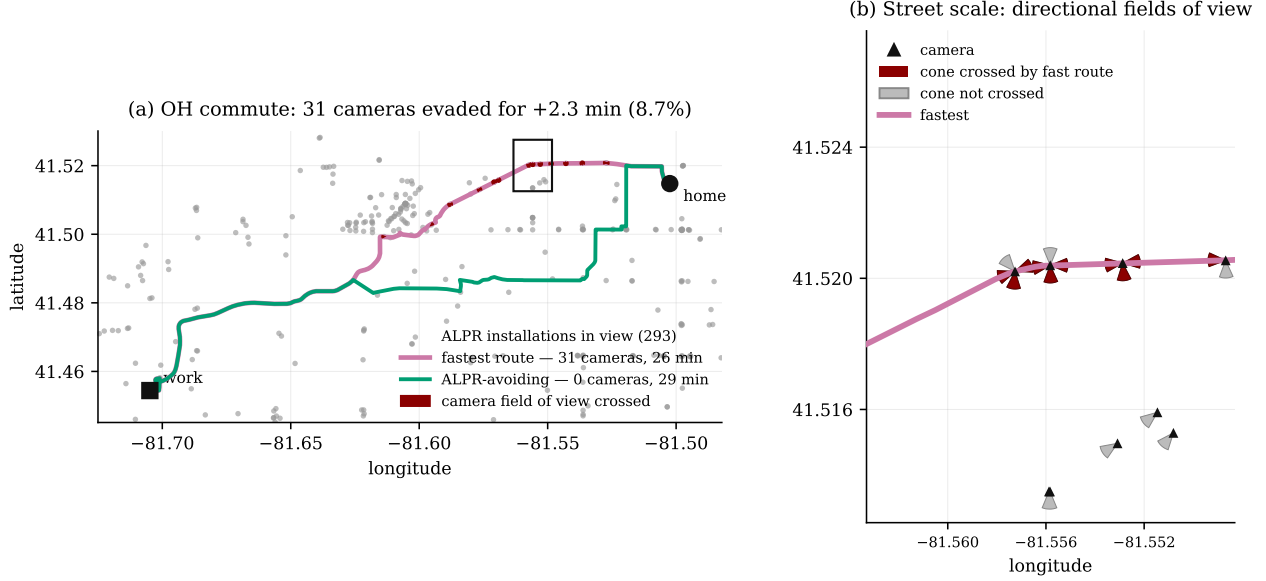


Figure 4: An illustrative high-exposure commute in the Cleveland area—chosen for legibility, not typicality. (a) The fastest route crosses 31 camera fields of view; the avoiding route crosses none, for 2.3 min (8.7%). (b) At street scale the directional model is visible: cones the fast route crosses are dark, cones aimed elsewhere are grey and cost nothing.

25.7 % overhead; in Illinois 90.6 % do, in California 86.7 %. At the other end 14.8 % of New Hampshire’s commuters pass one, 11.8 % of Montana’s, 5.4 % of Vermont’s, and none of Alaska’s sampled commuters. In fifteen jurisdictions the median extra time is zero: most of their commuters have nothing on the map to avoid.

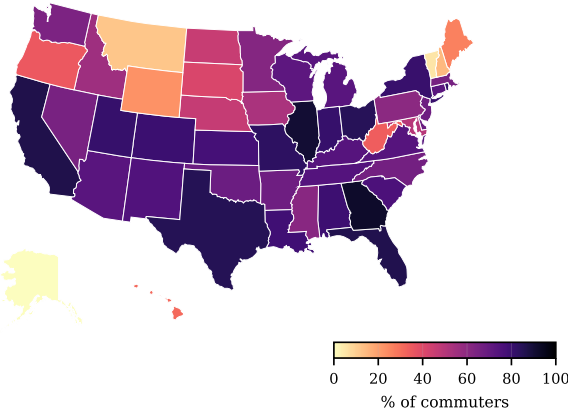
These are differences in the map at least as much as on the roads. Mapped density runs from 0.7 cameras per 100,000 residents in Alaska, and under five in Vermont, Maine and New Hampshire, to 92.8 in Georgia—a 7.1-fold spread between the 10th and 90th percentile states—and state exposure tracks it at $r = 0.82$. The departures from the line are what a statewide rate would produce: New York’s commuters pass more cameras than its statewide density predicts, because commuting concentrates where cameras do while the density is averaged over the whole state; Kansas’s pass fewer.

Per camera, evasion tends to be cheaper where cameras are denser. Across the 48 states with at least 100 evading commuters, the median price per camera falls as cameras per kilometre rise ($r = -0.29$ on a log scale), from 2.49 min in New Hampshire to 0.17 min in the District of Columbia. Dense grids offer a parallel street one block over; rural and intercity routes often have one road. Across states the relationship is weaker than the metropolitan spot-checks suggested—Georgia, the most densely watched large state, still pays 1.11 min a camera—so we state it as a tendency, not a law: the burden of refusal falls, if anything, hardest on drivers with the fewest cameras to refuse.

5.4 Demographic contrasts

Both ethnic and racial contrasts are statistically distinguishable nationally, and the Hispanic one is now the larger: commuters from the most Hispanic quartile of origin tracts pass 1.46 times as many cameras as those from the least. Income is not distinguishable in counts—but higher-income origin tracts show materially *denser* per-kilometre exposure (0.117, 4 against 0.084, 6) offset by shorter

(a) Share of commuters passing at least one ALPR



(b) Measured exposure tracks mapping intensity

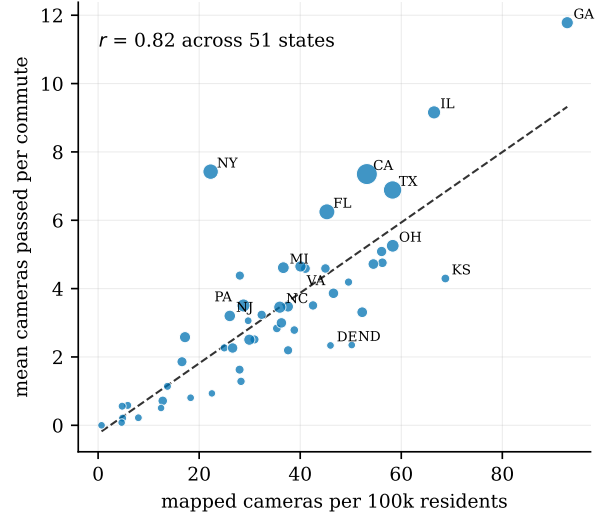


Figure 5: (a) Share of each state’s commuters whose unavoids route passes at least one mapped ALPR. (b) Mean cameras passed against mapped cameras per 100,000 residents, points sized by commuters. Mapping intensity alone tracks most of the variation between states ($r = 0.82$)—the central interpretive problem of Section 5.5.

Table 4: National quartile contrasts, commuter-weighted. Each ratio carries its own stratified-bootstrap interval; “disjoint” marks non-overlapping 95 % intervals on the two quartile means, the criterion used throughout.

Contrast	Ratio [95 % CI]	Cameras/km (Q1 vs Q4)	Verdict
Income Q1/Q4	0.96 [0.90–1.01]	0.084, 6 vs 0.117, 4	count overlaps; rate disjoint
% non-Hisp. Black Q4/Q1	1.39 [1.32–1.47]	0.073, 3 vs 0.123, 8	both disjoint
% Hispanic Q4/Q1	1.46 [1.39–1.54]	0.076, 9 vs 0.107, 4	both disjoint

commutes. Only reporting count and rate as peers surfaces that; a design treating the rate as a mere control would have discarded it.

5.5 Coverage-bias sensitivity, and why it governs the reading

Figure 7 is the most consequential figure in the paper. Recomputing quartiles *within each county* and pooling by commuters—so tracts are ranked only against neighbours subject to the same local mapping effort and broadly the same urban form—collapses every demographic gradient. The racial gradient falls from $1.39\times$ (disjoint intervals) to $1.06\times$ [0.99–1.13]; the Hispanic gradient, the largest nationally, from $1.46\times$ to $1.01\times$ [0.94–1.08]; income stays indistinguishable, its highest quartile against its lowest at $0.94\times$ [0.88–1.01]. Within-county per-kilometre rates are close: 0.116, 3 against 0.126, 8 by racial composition.

On the current map the demographic gradients are therefore **entirely a between-county phenomenon**—a statement that, as Section 5.7 shows, could not have been made of the sparser maps before it.

Mapped camera density varies enormously across the states—7.1-fold between the 10th and 90th percentile states, and far more at the extremes—and state exposure tracks it at $r = 0.82$ (Figure 5b). From any single map we cannot distinguish “Pennsylvania has few cameras” from “Pennsylvania

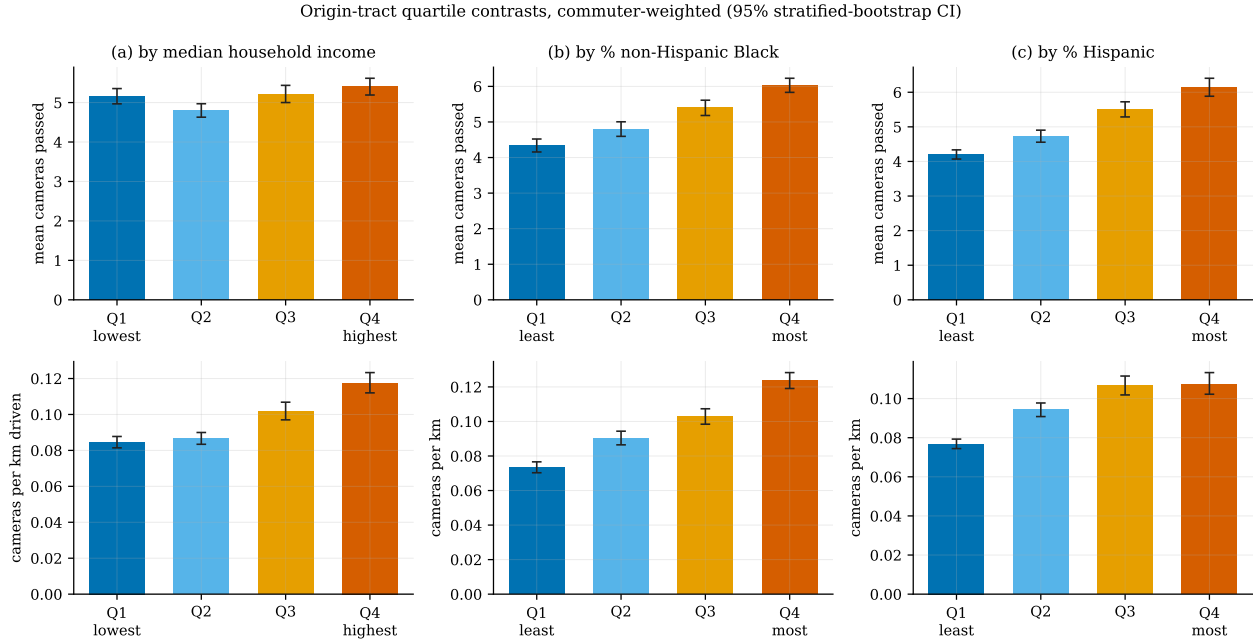


Figure 6: Origin-tract quartile contrasts, commuter-weighted, with 95% stratified-bootstrap intervals. Top row: mean cameras passed. Bottom row: cameras per kilometre driven. Income shows a per-km gradient with no count gradient—higher-income origins have denser but shorter commutes.

has few mappers”; the map’s history in Section 5.7 shows what can be said.

Three explanations remain consistent with these data, and this design separates none of them:

1. **Differential mapping effort.** Volunteer contributors are unevenly distributed, and where they are absent cameras are absent from the map but present on the pole.
2. **Urban form.** High-%Black and high-%Hispanic tracts are disproportionately urban and urban roads carry more cameras per kilometre, so the national contrast is substantially an urban/rural contrast.
3. **Genuine between-jurisdiction procurement differences.** Municipalities really do differ in what they buy.

We therefore decline to report a national demographic finding as a finding about surveillance siting. What we report is narrower and defensible: a gradient exists in the observable data, it is between-county, and it does not survive local ranking.

5.6 By camera vendor

Flock Safety holds 81.5% of mapped installations in the fifty states and the District but supplies only 67.8% of commuter exposure. Per unit deployed, non-Flock cameras generate **2.26×** the exposure (Table 5). The mechanism is visible in the cone geometry of Table 2: non-Flock units are multi-head more than six times as often, and DOT and tolling installations sit on the limited-access corridors commuters actually use, while a Flock unit watches one residential approach that fewer commutes cross.

This reframes the vendor’s dominance. Flock wins on count and loses on reach: its deployment model puts many cameras where comparatively few commuters pass, consistent with a product sold to municipalities and homeowners’ associations for neighbourhood coverage rather than to

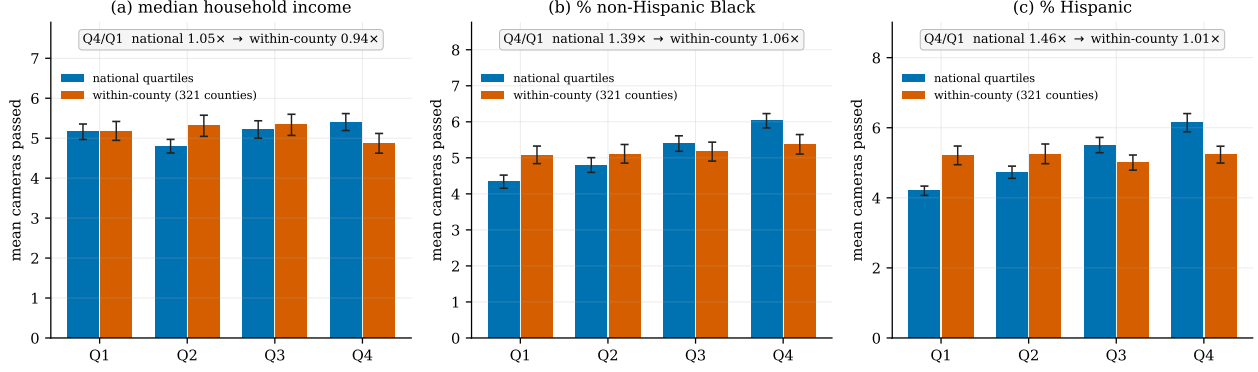


Figure 7: The central negative result. Ranking origin tracts within their own county rather than nationally collapses every demographic gradient to indistinguishability. 321 counties with $n \geq 40$ contribute 68 % of the sample and 63 % of its commuters.

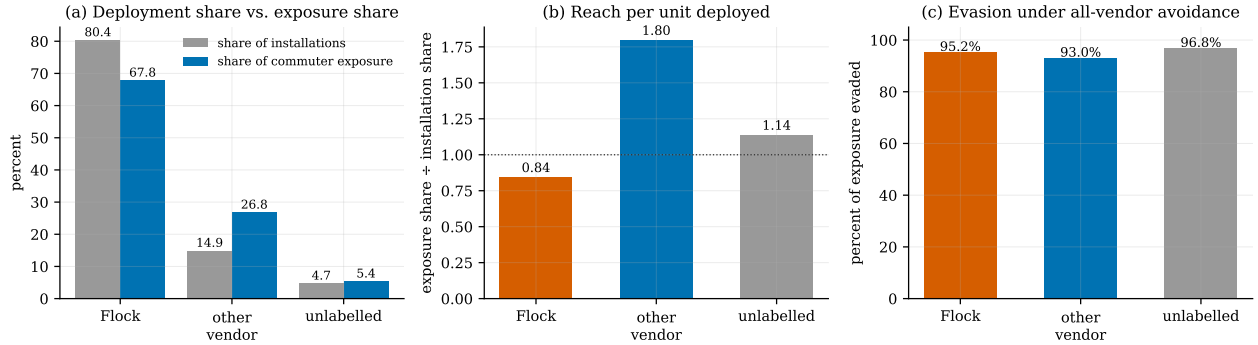


Figure 8: Vendor-resolved exposure, commuter-weighted. (a) Flock’s share of exposure falls well below its share of installations. (b) Reach per unit deployed: share of exposure divided by share of installations. (c) Both are avoided at comparable rates under the all-vendor model.

transport agencies for corridor coverage.

The gradient is stronger in transport infrastructure

Splitting Section 5.4 by vendor locates where the national disparity lives.

Both major camera populations show a distinguishable national gradient, but they are not equal: transport-agency and tolling cameras reach commuters from the highest quartile of Black population at 1.57 $\times$ the rate of the lowest, against 1.34 $\times$ for Flock—an excess of 57 % against 34 %, in infrastructure that no municipal council votes on.

A historical reading suggests itself. Mid-century urban highway construction routed limited-access corridors through Black neighbourhoods on a scale that is extensively documented, and tolling and DOT camera infrastructure sits on exactly those corridors. On that account the stronger gradient would be a decades-old artefact of road building rather than a contemporary artefact of camera procurement. We note the hypothesis; this design cannot test it.

And the discipline of Section 5.5 applies here too. Under within-county ranking Flock’s gradient vanishes (1.02 $\times$) and so does the untagged set’s (1.02 $\times$). The transport-agency gradient narrows from 1.57 $\times$ to 1.16 $\times$, and here the evidence is genuinely on the line. By the criterion used

Table 5: Vendor-resolved exposure, commuter-weighted over the 51-state sample. Installation shares are of the 139,391 cameras in the 50 states and the District of Columbia that contribute cones. Reach is share of exposure divided by share of installations: Flock holds four-fifths of installations and supplies two-thirds of exposure, and per unit deployed non-Flock cameras reach $2.26\times$ as many commuters.

Set	Installations	Share of exposure	Reach	Commuters ≥ 1	Evaded
Flock Safety	81.5 %	67.8 %	0.83	65.5 %	95.2 %
Other vendor	14.3 %	26.8 %	1.88	32.0 %	93.0 %
No vendor tag	4.2 %	5.4 %	1.27	11.6 %	96.8 %

Table 6: Racial composition contrast (% non-Hispanic Black, Q4/Q1) resolved by vendor and then subjected to the within-county control of Section 5.5, with stratified-bootstrap intervals on each ratio. Verdicts apply the criterion used throughout (non-overlapping intervals on the two quartile means); two ratio intervals exclude 1 although their quartile intervals overlap—the untagged national contrast, barely, and the transport-agency within-county contrast.

Camera set	National	Within-county	Verdict
Flock Safety	1.34 [1.26–1.42]	1.02 [0.94–1.11]	national only
Other vendor	1.57 [1.42–1.75]	1.16 [1.04–1.29]	national; within-county borderline
No vendor tag	1.18 [1.01–1.37]	1.02 [0.85–1.22]	neither
All vendors	1.39 [1.32–1.47]	1.06 [0.99–1.13]	national only

throughout it is not distinguishable—quartile means of 1.49 [1.39–1.60] against 1.72 [1.59–1.86]—yet the interval on the ratio itself, 1.04–1.29, excludes parity. Across the fifteen states of the study’s September round it passed the criterion narrowly ($1.20\times$); across all fifty-one it narrowly fails. We report a small within-county gradient in tolling and DOT exposure that the evidence neither establishes nor excludes, and that is consistent with the corridor reading above: a limited-access highway routed through a Black neighbourhood is an intra-county pattern in a way that a municipality’s purchasing decision is not. Holding RQ5 to a weaker evidentiary standard than RQ3 would let a between-county artefact return wearing a vendor label; holding it to the same standard leaves this one exactly on the line.

A correction, and what it cost. An earlier pass of this analysis ran on 15,043 of the 17,580 routes and reported that Flock exposure showed *no* racial gradient at all ($0.98\times$, overlapping; the July snapshot’s full-sample value was $1.20\times$, disjoint). That was wrong, and the error is instructive. The route-geometry sidecar is a gzip file appended to by more than one process, which makes it several concatenated gzip members; the reader used a single decompressor, which stops at the end of the first member and returns *successfully*. The missing 14 % was not merely noise—it moved Flock from “no gradient” to a disjoint gradient and would have supported a considerably stronger and more quotable claim than the data sustain. We record it because a silent truncation that returns success is the failure mode least likely to be caught by inspection, and because the corrected finding is the less convenient one.

Table 7: The same 56,131 commutes on the same roads against the camera map of each date, commuter-weighted. Exposure from re-scoring each saved route; avoidance from routing every commute exposed on that date, with the share reaching zero and the median extra time given for those commuters. The July 2026 snapshot was re-scored, not re-routed.

Map date	Mapped ALPR	Passing ≥ 1 (%) [95% CI]	Mean passed	Exposed who reach zero (%)	Extra min, exposed	Min per camera
2024-01-01	1,112	2.2 [2.0–2.4]	0.05	95.0	0.77	0.42
2024-07-01	1,420	3.3 [3.0–3.5]	0.08	96.0	1.00	0.49
2025-01-01	7,583	12.3 [11.9–12.7]	0.35	92.5	1.20	0.61
2025-07-01	22,690	29.4 [28.8–30.0]	0.94	92.4	1.36	0.65
2026-01-01	63,522	55.9 [55.3–56.4]	2.44	86.9	1.98	0.70
2026-07-24	120,272	71.7 [71.2–72.2]	4.40	—	—	—
2026-09-22	142,257	75.3 [74.8–75.7]	5.14	79.0	3.14	0.76

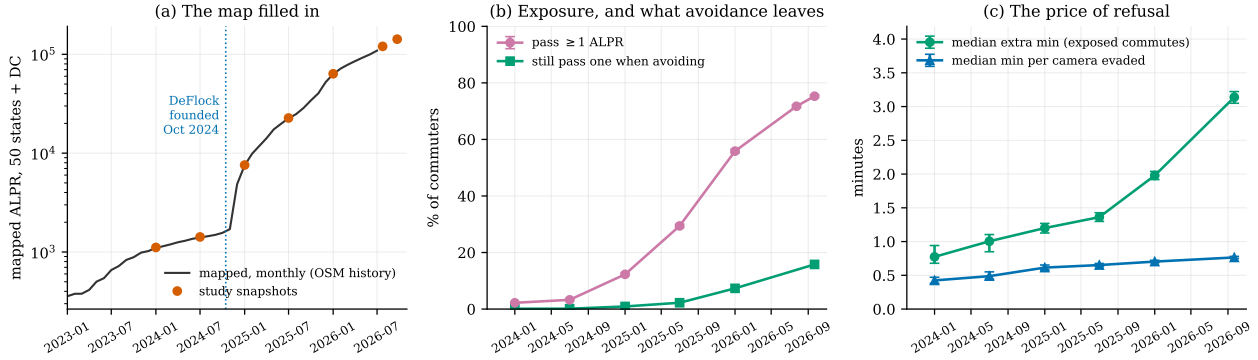


Figure 9: (a) Mapped ALPR in the fifty states and DC: monthly from the ohsome history API, and the study’s seven maps. (b) Share of commuters passing at least one mapped reader (95 % CIs), and the share still passing one on the avoiding route. (c) For commuters exposed on each date, the median extra minutes to avoid every camera, and the median minutes per camera evaded.

5.7 Thirty-three months of mapping

Everything above describes the map of 22 September 2026. We re-asked the same questions of the map as it stood on five earlier dates, reconstructed exactly from OpenStreetMap’s history (Section 3.3), and of the July 2026 snapshot: the same 56,131 commutes on the same roads, each saved route re-scored against each date’s cones, and every commute that passed a camera on a date routed again to avoid that date’s cameras (Section 4.4; the July snapshot was re-scored only). Table 7 and Figure 9 give the national picture, Figure 10 every state. Each date should be read as what had been *mapped* by then, not what had been installed.

The map filled in. On 1 January 2024 OpenStreetMap held 1,112 ALPR installations in the fifty states and the District of Columbia, four-fifths of them with no vendor tag, and twenty-six jurisdictions had none at all. Growth was slow until DeFlock was founded in October 2024 and abrupt after it: 7,583 by January 2025, 22,690 by July, 63,522 by January 2026 and 142,257 by September, a 128-fold increase in thirty-three months (Figure 9a). By July 2025 only Hawaii, Vermont and Alaska had none; by September 2026 every state had some. Flock’s share of the

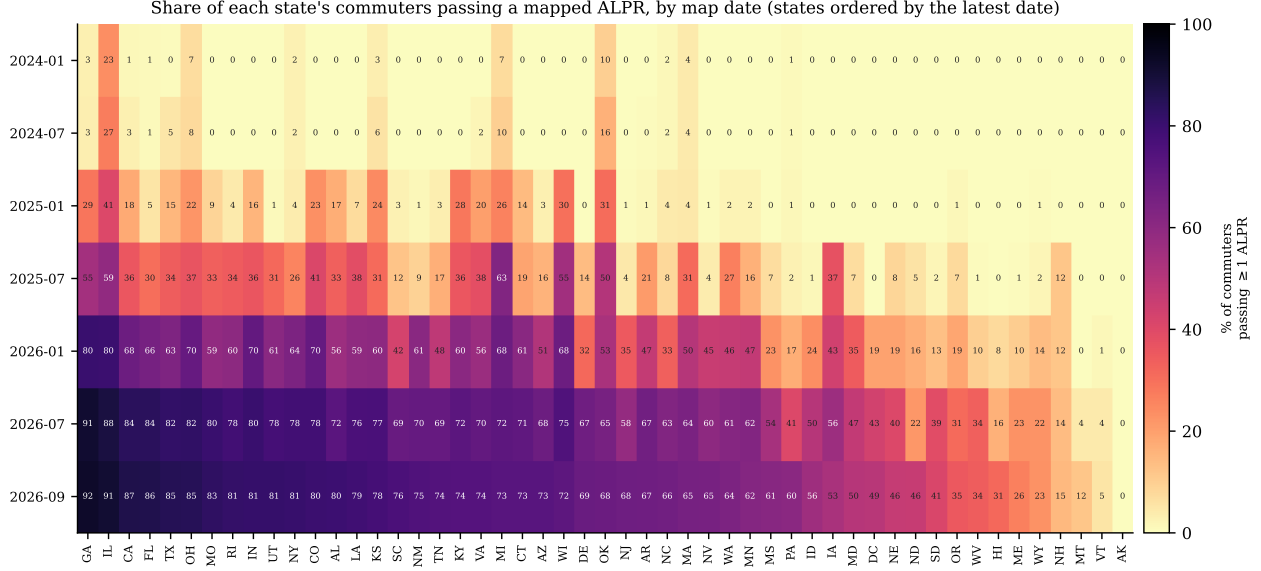


Figure 10: Share of each state’s commuters passing at least one mapped ALPR on each map, states ordered by the latest.

mapped network went from 17 % to 72 % during 2024 and has stayed near four-fifths since, while the untagged share fell from 81 % to 5 % and the named non-Flock vendors rose from 1 % to 15 % as contributors tagged them.

Exposure followed the map, almost in proportion. The share of commuters passing at least one mapped reader went from 2.2 % to 12.3 %, 29.4 %, 55.9 % and 75.3 %; the mean from 0.05 cameras to 5.14. Against the number of mapped cameras the elasticity is 0.92 across the seven maps. Per camera mapped, exposure has fallen by about a quarter since January 2024, from 0.049 to 0.036 encounters per commute per thousand cameras: the cameras mapped later lie on commute routes slightly less often than those mapped first, but not much less. Nor is there a sign of saturation. The last two months alone brought 18.3 % more mapped cameras and 17.0 % more encounters.

The states filled in at different times. In January 2024 Illinois was the only state where more than a tenth of commuters passed a mapped reader (23.5 %); by January 2025 fifteen were, by July 2025 thirty-one, by January 2026 forty-seven (Figure 10). In July 2025 Michigan’s commuters were the most exposed of any state’s (62.8 %), while Pennsylvania (2.4 %), Nevada (3.8 %), New Jersey (4.4 %) and North Carolina (8.2 %) would have looked nearly camera-free; all four are between 60 % and 68 % today. A state’s rank on the January 2025 map predicts its rank today only moderately (Spearman $\rho = 0.80$ on mean exposure; from January 2024, 0.54). Nor does the map only grow. Wisconsin and Iowa each gained cameras on net between July and September 2026 yet lost about three points of exposure: 190 and 123 of their July cameras left the map, deleted or retagged, and the cameras newly mapped added less exposure than was lost.

Refusal got dearer, and not only because there was more to refuse. On the sparse maps of 2024 a commuter who passed a mapped camera could almost always get around it: 95.0 % [92.7–96.9] of exposed commuters could reach zero, for a median of 0.77 min. On the September

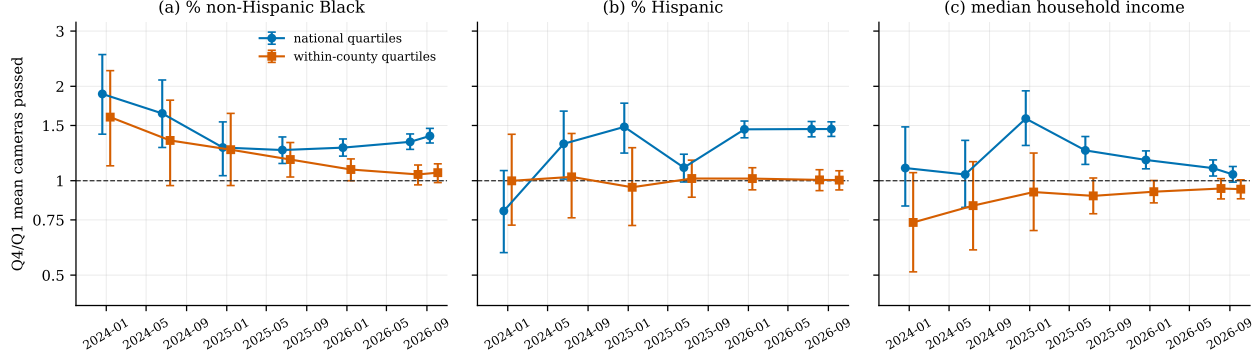


Figure 11: The demographic contrasts of Figure 7 on every map: mean cameras passed from the highest quartile of origin tracts over the lowest (for income, the richest over the poorest), with quartiles ranked nationally and within county, and 95 % CIs on the ratio. Log scale; the dashed line is parity.

2026 map 79.0 % [78.4–79.7] can, for a median of 3.14 min, having declined on every map since mid-2024 (Table 7; Figure 9c). Most of that rise is the number of cameras on each trip, but the price of evading each one rose too, from 0.42 min [0.38–0.47] to 0.76 min [0.75–0.78]. Two things moved it, and a fixed sample can separate them. One is who was exposed: the early maps covered dense metropolitan grids, where Section 5.3 finds evasion cheapest, and the later ones reached rural and intercity roads. The other happened on the same trips. Followed from the July 2025 map to September 2026’s, the 12,647 commutes exposed in July came to pass 9.9 cameras on average instead of 3.2; the share of them that can reach zero fell from 92.4 % to 72.6 %, and the price per camera rose from 0.65 min [0.63–0.68] to 0.81 min [0.79–0.84]. Every cohort followed for fifteen months or more shows the same—25 % to 36 % more per camera by September 2026, each interval clear of its starting one: as more of a trip’s cameras, and of its alternatives’, are mapped, each costs more to avoid. Over the two months between the 2026 snapshots the change is too small to see, which is why the fifteen-state comparison below finds the price per camera unchanged.

The national equity gradients moved with the map. Figure 11 repeats the contrasts of Sections 5.4 and 5.5 on every map. A study run on the January 2025 map would have reported that commuters from the richest quartile of tracts pass $1.58\times$ [1.30–1.93] as many cameras as those from the poorest; today the ratio is $1.05\times$ [0.99–1.11]. The Hispanic contrast was $0.80\times$ [0.59–1.08] in January 2024, $1.48\times$ [1.22–1.77] a year later, $1.10\times$ [0.99–1.21] in July 2025, and $1.46\times$ from January 2026 on. Each would have been reported with a confidence interval, three of the six quoted here excluding parity, and each describes at least in part where contributors had reached. Ranked within county, the Hispanic contrast lies between 0.95 and 1.03 on every map, and the income contrast between 0.74 and 0.94—at or below parity—on every map.

The Black contrast is the instructive exception. Nationally it fell from $1.89\times$ to $1.25\times$ by July 2025 and has risen since, to $1.39\times$. Within counties it was $1.60\times$ [1.12–2.24] on the January 2024 map—1,112 cameras, four-fifths without a vendor tag, past which 2.2 % of commuters drove—and it has fallen steadily since: 1.34 and 1.26 on the next two maps, 1.17 [1.03–1.32] in July 2025, 1.09 in January 2026, and 1.05 and 1.06 [0.99–1.13] on the two 2026 snapshots. On a sparse map the within-county control is not enough. It removes differences in mapping effort between counties, but when a county’s map holds a handful of cameras, which of its neighbourhoods they stand in is itself the choice of a few contributors, or reflects the few agencies whose cameras were mapped first. The check we recommend is therefore necessary rather than sufficient: it becomes informative as

the map within counties approaches completeness, and the convergence toward parity across seven maps is the best evidence available that the current map is nearer that point than any earlier one.

Fifteen states, fully re-routed. The trend’s last step can be checked against a comparison that re-routed everything. For the fifteen states of the study’s earlier rounds, the same 17,580 commutes were routed in full against both 2026 snapshots on the same road extract, and every baseline route was identical to the metre and the second. Camera encounters grew by 18.3 % against 18.3 % more mapped cameras, the median price of evading one stayed at 0.78 min to the second decimal, and Pennsylvania’s mean exposure rose 88 % in two months.

What follows. Three things. The absolute exposure figures of this paper are a lower bound that has risen with every map and will keep rising; any study built on crowdsourced surveillance data dates, and should say when its map was drawn. The price of refusal dates as well: it has risen with every map, per camera as well as in total, and today’s figures describe a map that is still filling in. And no national equity gradient measured from these data is a finding about cameras until it has survived both the within-county control and the question of how complete the map was when it was measured. Over thirty-three months the national gradients rose, fell and changed direction, while the within-county ones held at parity or converged toward it.

6 Discussion

6.1 Refusal is cheap, and that is the policy-relevant fact

The strongest result here is also the simplest: for 84.2 % of American commuters, complete avoidance of licence-plate surveillance costs a median of 1.62 min. Arguments for ALPR deployment often rest implicitly on ubiquity—the idea that the network is comprehensive enough that avoidance is futile and therefore the privacy loss is unavoidable rather than chosen. At current mapped density that premise is false for five commuters in six. It has been false on every map since 2024, but it is becoming less so: as the map filled in, the share of exposed commuters who can reach zero fell from 95 % to 79 %, and the median cost for those with a camera to avoid quadrupled (Section 5.7).

The corollary cuts the other way. If avoidance is this cheap, it is also cheap for people with reason to evade legitimate investigation, which bounds how much investigative value a fixed-camera network can deliver against a motivated subject while it continues to read every unmotivated one.

6.2 The equity question is not answerable from crowdsourced data

We set out to measure whether surveillance burden falls unevenly, and we can report only that the question is harder than it appears. The national gradients are real in the data, substantial ($1.39\times$ by Black and $1.46\times$ by Hispanic share of the origin tract), and statistically clean—and they evaporate under a control that any careful reviewer would demand. Of every camera population, only transport-agency cameras retain a within-county gradient, small, with evidence that sits exactly on the line.

The map’s own history makes the point without any control at all. Run on the map of January 2025, this study would have found that the richest quartile of tracts passes $1.58\times$ as many cameras as the poorest; run six months later, that the Hispanic gradient had shrunk to $1.10\times$ from $1.48\times$; run today, that there is no income gradient and that the Hispanic one is back at $1.46\times$. Each

answer would have carried a clean confidence interval, and each was a photograph of a mapping effort in progress (Section 5.7).

We think this generalises. A considerable amount of accessible research on surveillance infrastructure necessarily rests on volunteer-mapped data, because no authoritative national inventory exists. Any such study that reports a demographic correlation without a within-locality control is at serious risk of reporting the demography of *mappers* rather than of the surveilled, and a study that reports one from a sparse map is at risk even with the control. We would encourage the control to become routine, together with the date of the map and, where the edit history allows, a check of how the result moves as the map fills in.

6.3 Attention and exposure are aimed at different infrastructure

Flock Safety is the name in municipal council debates, in local journalism, and in the activist mapping effort that produced the data underlying this paper. It holds four-fifths of mapped installations. Yet per unit deployed its cameras reach fewer than half as many commuters as a tolling or transport-agency camera, and the racial gradient in its exposure is weaker ($1.34\times$ against $1.57\times$).

We do not read this as exonerating, for a reason the measurement itself makes clear. A camera at the entrance of a residential subdivision does something qualitatively different from a camera on a toll gantry: it observes a small population repeatedly rather than a large population once, and repeated observation of a known small set is precisely what makes pattern-of-life inference possible. Commute exposure measures breadth, not depth, and Flock’s model appears optimised for depth. A study designed around repeat observation of the same vehicle would likely reverse the ordering reported here, and we would expect it to.

What the result does establish is that a distributional concern has been attached disproportionately to one vendor. The infrastructure with the steeper measured gradient is installed by transport agencies, largely absent from the municipal debate, and sited on corridors laid down decades before ALPR existed—and it is the one gradient whose within-county ratio still excludes parity once tracts are compared only with their neighbours. Regulation aimed only at the vendor currently in the news would leave the larger measured disparity untouched.

6.4 Where the burden actually concentrates

The distributional finding that best survives the coverage critique is geographic rather than demographic, and it is counter-intuitive: per camera, avoidance tends to be cheapest in dense metropolitan grids and dearest in rural and intercity travel. Network topology, not camera count alone, sets the price of refusal. The tendency is modest across whole states (Section 5.3), but it is robust to the coverage critique in a way the demographic results are not, because it rests on road-network structure—which OSM maps close to completely—rather than on camera counts, which it does not.

7 Limitations

7.1 Tract centroids are not addresses

Every route runs centroid to centroid; real commutes begin in a driveway. In large or irregular tracts the centroid can sit far from any residence. This adds noise in both directions and is the main reason we report distributions rather than point estimates. It also costs coverage: where a

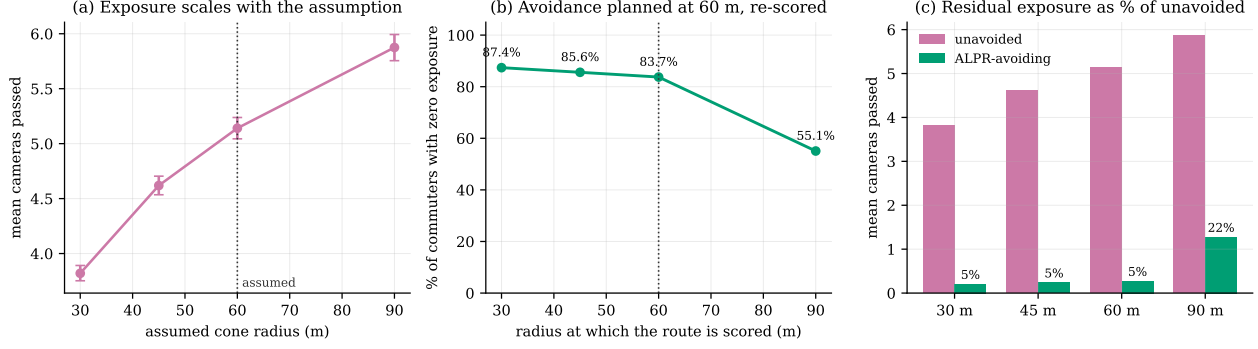


Figure 12: Sensitivity to the assumed cone radius. (a) Measured exposure scales gently— halving the radius removes only a quarter of it. (b) The share of commuters reaching zero exposure is flat below the assumed value and falls sharply above it. (c) Residual exposure as a fraction of unavoided, by radius.

tract’s internal point lies too far from any road to snap to, the commute cannot be routed at all, which removes a fifth or more of the sample in Alaska, Hawaii, Montana and Wyoming (Section 4.6).

The stricter caution: tract demographics describe tracts, not the people routed. “Commutes originating in high-poverty tracts pass more cameras” is supportable; “poor people pass more cameras” is not.

7.2 Crowdsourced camera data is non-randomly missing

Absolute exposure figures are a **lower bound**: where contributors are absent, cameras are absent from the map but present on the pole. The hazard is not undercounting but its correlation with the variables of interest, quantified in Section 5.5. Every demographic statement in this paper is conditional on that, and the condition is not a formality.

7.3 Cone geometry: swept, and one result depends on it

The 60 m radius was chosen by reasoning—a pole set back from the kerb needs a cone long enough to reach the carriageway it watches—and the 45° default span from the 82 % of explicit arcs that span exactly that. Neither was measured. We therefore swept the radius over {30, 45, 60, 90} m, rebuilding the cone geometry at each and re-scoring all 56,131 routes (Figure 12).

Table 8: Cone-radius sensitivity, commuter-weighted. Exposure is mildly sensitive to the assumption; the avoidance result is not, until the radius grows by half. At the assumed 60 m the clean share is 83.7 % rather than the 84.2 % of Table 3 because the sweep re-scores stored route geometry (Section 4.4).

Radius	Mean cameras	vs 60 m	Avoided routes still clean	Residual
30 m	3.82	0.74×	87.4 %	5.4 %
45 m	4.62	0.90×	85.6 %	5.2 %
60 m	5.14	assumed	83.7 %	5.3 %
90 m	5.88	1.14×	55.1 %	21.7 %

Two things follow, and they point in opposite directions.

Exposure is robust. Halving the radius removes only a quarter of the measured exposure, and 90 m adds 14 %. Cones are small relative to the road network at any of these values, so Section 5.1

does not rest on the choice.

The avoidance headline does depend on it. Routes planned against 60 m cones remain clean when re-scored at 45 m and 30 m—avoidance is not finely tuned to the assumption from below. But at 90 m the share reaching zero falls from 83.7 % to 55.1 %, a drop of 28.7 points (95 % CI $[-29.3, -28.1]$), and residual exposure quadruples from 5.3 % to 21.7 % of baseline. If real fields of view are half again as long as we assume, more than a third of the routes this paper counts as clean are not.

That figure is nonetheless a *lower* bound rather than an estimate, for a reason worth being precise about. These routes were *planned* against 60 m cones and only *re-scored* at 90 m; a router given 90 m cones would find different clean routes, and the denser the network the more of them there are. Measuring the true value requires a complete re-route at each radius—most of a day apiece at this sample’s size—which we have not done. What the sweep establishes is that the avoidance claim is conditional on the radius in a way the exposure claim is not, and that the conditionality bites only above the assumed value.

7.4 A single avoidance strength

All results use the maximum strength. Intermediate settings are the practically interesting ones for a shipped application and they behave badly: the usable range is roughly $0.3 \rightarrow 0.01$ and the dead zone is *density-dependent*. A setting of 0.3 does nothing in Delaware while avoiding 62.5 % of cameras in Atlanta. A single fixed slider-to-multiplier mapping is therefore not viable; the mapping must adapt to local density. This is an open problem for the application and an unmeasured dimension of this paper.

7.5 Sample scope

All fifty states and the District of Columbia, but within-state commutes only (`od_main`), so cross-border metropolitan areas—New York/New Jersey, Philadelphia, Washington, Kansas City, St Louis, Portland—are systematically truncated at the state line. Several are exactly the dense multi-jurisdiction areas where the question is most pointed. The District is the extreme case: most people who work there live in Maryland or Virginia, so its sample describes only commutes with both ends inside the District. Alaska’s flows are from 2016 and Michigan’s from 2021, the latest LODS publishes; the other 49 jurisdictions’ are from 2022.

7.6 Past maps record mapping, not deployment

The trend of Section 5.7 is a trend in what had been mapped. OpenStreetMap’s installation dates are too sparse to separate the two—47 of 142,991 nodes carry one—so a camera first mapped in 2026 may have stood since 2021, and the 128-fold growth of the map cannot be read as a rate of deployment. Absolute figures date accordingly. The relative claims are the ones to carry forward, and across seven maps they held: the cheapness of refusal—under a minute per camera evaded on every map, though rising—the within-county collapse of the national gradients on every map dense enough to test it, and the instability of the national gradients themselves. What the maps cannot establish is that the current one is complete. The convergence of the within-county ratios toward parity is what a true effect of zero would produce, and also what a real effect would produce if it were concentrated in counties still being mapped. The transport-agency gradient of Section 5.6—narrowly distinguishable across the fifteen states of the September round, narrowly not across all fifty-one—is where that distinction matters most, and a vendor-resolved trend would test it.

7.7 Free-flow travel times

Routing uses free-flow speeds with no traffic, time-of-day, or turn-restriction penalties, so both travel times are optimistic. The *difference* between them—the quantity actually reported—is more robust than either absolute, since both share the bias.

7.8 Avoidance is a preference, not a guarantee

`multiply_by: 0.01` makes observed roads expensive, not impassable. Where no alternative exists the router correctly uses the watched road. “Zero cameras” means the optimiser found a clean path, not that one was guaranteed to exist.

8 Reproducibility

Fixed seed 20260725 throughout, for both sampling and bootstrap. All external data sources are public and require no API key.

Component	Path
Data acquisition	<code>research/data/fetch.sh</code>
Camera fetch (continental)	<code>server/alpr/fetch-na-cones.sh</code>
Cone construction	<code>server/alpr/build_cones.py</code>
Past camera maps (Overpass)	<code>server/alpr/fetch-history.sh</code>
Past camera maps (history dump)	<code>server/alpr/history_from_planet.py</code>
Monthly mapping history	<code>research/scripts/fetch_mapping_history.py</code>
Cone-set variants	<code>server/alpr/build-variants.sh</code>
Snapshot statistics	<code>research/scripts/snapshot_stats.py</code>
Routing configuration	<code>server/graphhopper/config/trames.yml</code>
Graph build	<code>server/graphhopper/rebuild-with-cones.sh</code>
Commute sampling	<code>research/scripts/build_sample.py</code>
Experiment	<code>research/scripts/run_experiment.py</code>
Routing on past maps	<code>research/scripts/run-history.sh</code>
Commuter weights, bootstrap	<code>research/scripts/wstats.py</code>
Trend	<code>research/scripts/trend.py</code>
Geometry backfill	<code>research/scripts/backfill_geometry.py</code>
Vendor rescoring	<code>research/scripts/vendor_exposure.py</code>
Analysis	<code>research/scripts/analyze.py</code>
Post-processing, one command	<code>research/scripts/run-analyses.sh</code>
Snapshot comparison	<code>research/scripts/compare_snapshots.py</code>
Figures	<code>research/paper/make_figures.py</code>
Tables	<code>research/paper/make_tables.py</code>

The 18 GB OSM extract, the 14 GB graph caches and the 163 GB OSM full-history dump are not redistributed; the acquisition and import scripts rebuild all of them. Commuter weights come from `research/out/sampling_frame.json` and `sample_draws.csv`, written by the sampler.

A Every state

Table 9: Every state and the District of Columbia. Commuters are in-state commuters in the LODES frame; “routed” excludes the unroutable commutes of Section 4.6. Mapped ALPR are nodes inside the state on 2026-09-22, per 100,000 residents from ACS 2022. Exposure and avoidance figures are within-state estimates; the national row is commuter-weighted as in Table 3.

State	Commuters (M)	Commutes routed	Mapped ALPR	per 100k	Mean passed	≥ 1 (%)	Zero when avoiding (%)	Extra min	Over- head (%)
Alabama	1.79	1,159	2,829	56.3	4.76	79.6	82.4	2.83	9.1
Alaska	0.26	608	5	0.7	0.00	0.0	100.0	0.00	0.0
Arizona	2.88	1,150	2,696	37.6	3.47	72.7	93.2	1.10	4.6
Arkansas	1.06	1,146	1,068	35.4	2.84	67.4	92.6	1.09	3.7
California	16.82	1,171	20,932	53.2	7.35	86.7	80.8	2.40	8.8
Colorado	2.54	1,117	2,688	46.6	3.86	79.8	89.2	1.77	7.4
Connecticut	1.41	1,183	1,013	28.1	4.38	72.8	93.2	1.34	6.5
Delaware	0.34	1,104	457	46.0	2.34	68.7	91.7	0.81	4.1
District of Columbia	0.20	1,032	151	22.5	0.93	49.4	92.6	0.00	0.0
Florida	8.80	1,170	9,797	45.3	6.25	86.3	71.3	2.63	9.9
Georgia	4.19	1,182	9,956	92.8	11.78	91.9	54.3	7.78	25.7
Hawaii	0.55	835	85	5.9	0.58	31.5	95.9	0.00	0.0
Idaho	0.69	1,007	463	25.0	2.27	55.6	94.9	0.25	1.2
Illinois	5.29	1,188	8,483	66.5	9.16	90.6	90.5	3.37	15.1
Indiana	2.72	1,176	3,696	54.5	4.72	81.0	92.7	3.16	13.3
Iowa	1.27	1,150	893	28.0	1.63	53.0	96.7	0.00	0.0
Kansas	1.08	1,166	2,018	68.7	4.30	77.6	89.7	1.77	8.9
Kentucky	1.61	1,160	1,915	42.5	3.51	73.6	87.6	1.67	5.7
Louisiana	1.68	1,116	2,088	45.0	4.59	78.7	68.2	3.55	14.1
Maine	0.52	1,066	65	4.8	0.56	26.2	99.0	0.00	0.0
Maryland	2.13	1,183	1,023	16.6	1.86	49.7	90.3	0.00	0.0
Massachusetts	3.12	1,187	1,201	17.2	2.58	65.4	91.5	0.61	2.9
Michigan	3.79	1,190	3,687	36.7	4.61	72.8	94.7	1.70	6.5
Minnesota	2.49	1,163	1,516	26.6	2.26	62.0	95.3	0.39	1.5
Mississippi	0.97	1,125	1,149	38.8	2.79	61.3	89.4	0.67	1.8
Missouri	2.39	1,182	3,453	56.1	5.09	83.0	81.3	2.68	11.0
Montana	0.40	789	87	8.0	0.22	11.8	99.9	0.00	0.0
Nebraska	0.80	1,125	554	28.3	1.29	46.2	98.7	0.00	0.0
Nevada	1.33	1,132	961	30.9	2.51	65.2	95.4	0.46	2.4
New Hampshire	0.52	1,105	67	4.9	0.21	14.8	100.0	0.00	0.0
New Jersey	3.53	1,184	2,410	26.1	3.20	67.5	92.4	0.76	3.1
New Mexico	0.72	994	1,047	49.6	4.19	74.6	83.2	1.41	6.9
New York	8.09	1,194	4,452	22.3	7.42	80.8	62.8	2.42	12.2
North Carolina	4.21	1,179	3,765	36.0	3.46	66.4	88.2	0.94	2.7
North Dakota	0.30	977	390	50.2	2.35	45.9	99.5	0.00	0.0
Ohio	4.84	1,195	6,866	58.3	5.25	84.7	81.9	2.81	10.9
Oklahoma	1.43	1,168	1,493	37.6	2.19	68.4	96.1	0.41	1.1
Oregon	1.68	1,133	541	12.8	0.72	35.4	98.9	0.00	0.0
Pennsylvania	5.16	1,178	3,737	28.8	3.51	60.2	94.9	0.37	1.3
Rhode Island	0.38	1,142	325	29.7	3.06	81.4	83.7	1.46	10.5
South Carolina	1.93	1,159	2,110	41.0	4.59	76.0	83.1	2.12	7.0
South Dakota	0.33	906	163	18.3	0.81	41.2	99.7	0.00	0.0
Tennessee	2.75	1,182	3,619	52.3	3.31	73.9	85.3	1.64	4.3
Texas	12.37	1,176	17,043	58.3	6.89	84.9	81.2	2.26	10.1
Utah	1.49	1,091	1,063	32.4	3.23	81.1	95.7	1.71	8.4
Vermont	0.23	940	30	4.7	0.08	5.4	100.0	0.00	0.0
Virginia	3.30	1,171	3,454	40.0	4.65	73.5	86.1	1.15	4.5
Washington	3.19	1,145	2,301	29.9	2.50	63.9	92.1	0.49	1.6
West Virginia	0.52	1,118	246	13.7	1.14	34.0	93.4	0.00	0.0
Wisconsin	2.53	1,176	2,134	36.3	3.00	71.7	92.9	1.52	6.9
Wyoming	0.20	656	72	12.5	0.51	23.0	92.7	0.00	0.0
United States	132.8	56,131	142,257	25	5.14	75.3	84.2	1.62	6.7